

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Artificial Intelligence and Data Science) SEMESTER - 3 Summer 2025 (Supply.)

Course :Bachelor of Technology (Artificial Intelligence and Data Science) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: BTAIC303 - DATA STRUCTURE AND ALGORITHM USING PYTHON

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 How is a code block indicated in Python?
a. Brackets b. Indentation c. Key d. None
- What will be the output of the following code ?
a=3 b=1
print(a,b)
a,b=b,a
print(a,b)
a. 3 1 1 3 b. 3 1 3 1 c. 1 3 1 3 d. 1 3 3 1
- 3 Which of the following blocks will always be executed whether an exception is encountered or not in a program?
a. try b. except c. finally d. catch
- 4 What is fruitful function in Python?
a. function which will return value b. function which will not return value
c. function which will send value. d. none
- 5 What will be the Postfix expression for $(a+b+c)*d$
a. $ab+c+d*$ b. $ab+cd+*$ c. $abc++cd*$ d. None
- 6 Data Structure is Classified into :
a. Linear b. Non-Linear c. Non-Primitive d. All of these
- 7 What is a hash table?
a. A structure that maps values to keys b. A structure that maps keys to values
c. A structure used for storage d. A structure used to implement stack and queue
- 8 Which of the following is not a collision resolution technique?
a. Separate chaining b. Linear probing c. Quadratic probing d. Hashing
- 9 For a binary search algorithm to work, it is necessary that the array(list) must be
a. Sorted b. Unsorted c. In Heap d. Popped out of stack

- 10 Binary Search can be categorized into which of the following?
- Brute Force technique
 - Divide-and conquer
 - Greedy method
 - Dynamic programming
- 11 Which sorting is also called divide and conquer sorting?
- Quick sort
 - Bubble sort
 - Merge sort
 - Both a & c
- 12 What will be the output of the following code.
- ```
dict1 = {'first' : 'sunday', 'second' : 'monday'}
dict2={1:3, 2:4}
dict1.update(dict2)
print(dict1)
```
- { 'first':'sunday','second':'monday',1:3, 2:4}
  - { 'first':'sunday','second':'monday'}
  - {1: 3, 2: 4}
  - None of these
- Q2. Solve the following.
- Explain features of Python in detail. 6
  - Explain concept of stack with its operations. 6
- Q3. Solve the following.
- Define tree data structure. Explain different types of tree with examples 6
  - Write an algorithm for bubble sort? Sort following elements with bubble sort  
45,67,23,13,20,10 6
- Q4. Solve Any Two of the following.
- Explain concept of Inheritance and its types in python with syntax. 6
  - Construct a Binary search tree for following data and perform inorder, preorder and postorder 20,3,13,16,30,56,10,24,56 6
  - Write a python program to search element using Binary search method 6
- Q5. Solve Any Two of the following.
- Write an algorithm for quick sort and insertion sort method 6
  - Write a python program to construct BST and perform Preorder tree traversal. 6
  - Explain following a. Abstraction b. encapsulation c. class 6
- Q6. Solve Any Two of the following.
- Convert following infix expression to postfix :a.  $(x+y)^2 + (a+b)^2$  b.  $(p + q - r) * (s + t)$  6
  - Explain Linked list and its types with representation. 6
  - Write a python program to perform arithmetic operations using function. 6

\*\*\* End \*\*\*